KRISHNA SAI PENTAKOTA

Forward Deployed Engineer · AI & Data Engineer

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SUMMARY

Sole AI and data engineer at a ribbon manufacturer, reporting directly to the Managing Director. I embed with weaving, warehouse, sales and accounts teams, audit their messy Excel and ERP data, and ship validated PostgreSQL and Streamlit systems they can act on. My forensic audit of eight months of weaving production data showed that 82% of the plant's output shortfall came from machine downtime, not from targets set too high. I build with Python, SQL and AI coding tools (Claude, ChatGPT), with automated tests and reconciliation behind every number.

EXPERIENCE

AI Engineer, MISO Ribbon

Aug 2026 – Present

Atchuthapuram, Andhra Pradesh

- Ran a 20-day forensic audit of eight months of weaving production records (70 looms, Jan–Aug 2026), rebuilding wide-format monthly workbooks into 29,000+ machine-shift rows reconciled to the department's own totals (August within 21 yards on 6.6 million).
- Found errors hiding production from the books: formulas broken by copy-paste (one loom lost 13 days and 59,747 yards in a month), mislabelled date columns, swapped headers, rejection figures that disagreed between tabs. Logged every correction in an audit trail.
- Showed that 13.5% of the plant's target was charged to stopped machines; added machine-status rules and proved the shortfall was 82% downtime and 18% target-setting. Priced changeovers for the first time: 87 events, about 477,000 yards lost.
- Turned 1,064 production WhatsApp messages into a machine-stoppage log (254 events, 9 breakdown reasons) and redesigned the paper production log in English and Telugu to capture stop time, restart time and reason code.
- Built the weaving dashboard on PostgreSQL and Streamlit: constrained star schema, idempotent Excel ETL, 96 automated tests, 10 screens including a Telugu floor board, every query under 0.5 s.
- Delivered a sales and marketing dashboard (customer- and product-wise sales, open order book, on-time delivery) that reconciles against the accounts report; its PostgreSQL and FastAPI backend is in progress.
- Mapped the raw-material and finished-goods stores with the store teams and delivered a prototype 3D warehouse map and a store inventory system plan within the MD's one-month deadline.
- Ran a company-wide AI usage and hardware assessment and recommended AI seats and PC upgrades to the MD.

Content Associate, Oliveboard

Feb 2025 – Jun 2025

Bengaluru

- Created and reviewed technical and aptitude questions, solutions and explanations, with a focus on accuracy and consistency.
- Analysed learner feedback and performance trends to find and fix weak or confusing content.
- Started using LLMs (ChatGPT, Claude) in daily work: drafting question variants, cross-checking solutions and structuring explanations, with every AI draft reviewed before publishing.
- Set a review step for every question before it was published: one correct answer, clear wording and a worked explanation, checked against the same standard each time.

Junior Data Analyst, Tata Consultancy Services (client: Microsoft)

Feb 2022 – Dec 2023

Hyderabad

- Owned the quality gate for high-volume Microsoft client records: each batch was checked against the agreed standard, and discrepancies were logged with their cause before any figure reached a report.
- Turned recurring errors into fixes: traced them to where they started, passed the pattern to team leads with a worked example, and tracked the results in Excel reports so the team could see whether they were falling.
- Validated and verified client project records and documents against defined quality standards, keeping data accurate and audit-ready.
- Ran quality checks on high-volume records and flagged discrepancies for correction before they reached reports.
- Built and maintained Excel reports tracking operational activity and performance metrics for team leads.

SELECTED PROJECTS

Sri Ratna's Homoeo Clinic — website & booking

Delivered

Live website for a two-branch homoeopathy clinic on its own domain, with online hourly-slot booking stored in Postgres, a staff dashboard and a waiting-room TV view. The WhatsApp booking bot is built and goes live once Meta approves the business.

TimberTrack — timber dispatch and quality app

In progress

A trader trucks mango and teak logs from Andhra Pradesh to buyers in Rajasthan. TimberTrack measures and prices each log, records the load before it leaves, alerts the buyer on WhatsApp, and keeps one ledger per consignment.

SKILLS

Data and engineering	Python, SQL, PostgreSQL (schemas, constraints, views, roles), ETL, data validation and reconciliation, star-schema modelling, automated testing
Apps and BI	Streamlit, Plotly, HTML dashboards, advanced Excel
AI	Claude and ChatGPT for development and analysis, AI-adoption assessment, learning FastAPI, Pydantic and Docker
Domain	weaving and textile manufacturing, production efficiency and downtime, warehouse inventory, sales reporting, Kingdee EAS ERP data

EDUCATION

B.Tech, Civil Engineering, Vignan Institute of Information Technology, Visakhapatnam

2017 – 2021